

Design and Analysis of Algorithms CSEL30503MJ

Module - 3
Branch and Bound

Content

- Introduction
- Traveling Salesman Problem
 - Lower Bound Theory
 - Suggestive Questions

Introduction: Branch and Bound

Branch and Bound is a general algorithm paradigm used to find optimal solutions to optimization problems, especially integer programming and combinatorial problems like the Traveling Salesman Problem (TSP). It systematically explores a state-space tree of possible solutions by intelligently searching and pruning branches.

Key Concepts

- **Branching:** This involves dividing a large problem into smaller subproblems, effectively creating a search tree. Each node in the tree represents a subproblem.
- **Bounding:** This is the core of the algorithm. A **lower bound** is calculated for each subproblem. The lower bound is the minimum possible cost for any solution within that subproblem's branch.
- **Pruning:** If the lower bound of a subproblem is greater than the best solution found so far (the upper bound), that entire branch is discarded (pruned), as it cannot possibly contain a better solution. This significantly reduces the search space.

Traveling Salesman Problem (TSP)

The TSP seeks the shortest possible tour that visits each city exactly once and returns to the origin city.

- **Problem:** Find the minimum cost tour for cities A, B, C, D.
- **Branching:** The tree is formed by considering all possible paths. The first level branches on which city to visit after the start city.
- **Bounding:** A simple lower bound can be the sum of the costs of the cheapest edges connected to each vertex. For a partial tour like A → B, the lower bound is the cost of A → B plus the cheapest remaining edges to complete the tour.
- **Pruning:** If a partial tour's lower bound (e.g., A → B → C has a cost of 15) is already greater than a complete tour found earlier (e.g., A → C → D → B → A has a cost of 12), the entire A → B → C branch can be pruned.

Example: Traveling Salesman Problem

Problem 1: A salesman must visit four cities A, B, C, and D. The costs of travel between them are given by the following matrix. Find the optimal tour using the Branch and Bound method.

$$C = \begin{pmatrix} \infty & 10 & 15 & 20 \\ 10 & \infty & 35 & 25 \\ 15 & 35 & \infty & 30 \\ 20 & 25 & 30 & \infty \end{pmatrix}$$

Solution:

The solution involves reducing the cost matrix by subtracting the smallest element from each row and column. This process gives us a lower bound for the cost.

Example: Traveling Salesman Problem

Step 1: Row Reduction

- Subtract 10 from row A: (∞ 0 5 10)
- Subtract 10 from row B: (0 ∞ 25 15)
- Subtract 15 from row C: (0 20 ∞ 15)
- Subtract 20 from row D: (0 5 10 ∞)

1. Row-reduced matrix: $C1 = \begin{pmatrix} \infty & 0 & 5 & 10 \\ 0 & \infty & 25 & 15 \\ 0 & 20 & \infty & 15 \\ 0 & 5 & 10 & \infty \end{pmatrix}$

2. Lower Bound (LB): $10+10+15+20=55$.

Example: Traveling Salesman Problem

Step 2: Column Reduction

The first two columns already have a zero.

Subtract 5 from column 3: $\begin{pmatrix} 0 \\ 20 \\ \infty \\ 5 \end{pmatrix}$

Subtract 10 from column 4: $\begin{pmatrix} 0 \\ 5 \\ 5 \\ \infty \end{pmatrix}$

Final reduced matrix: C2=

$$\begin{pmatrix} \infty & 0 & 0 & 0 \\ 0 & \infty & 20 & 5 \\ 0 & 20 & \infty & 5 \\ 0 & 5 & 5 & \infty \end{pmatrix}$$

Total LB: 55+5+5=65.

Now we branch from this state. Let's choose the edge (A,B) with cost 10. The lower bound for this path will be 65+0=65. We can proceed by systematically exploring branches, pruning any that exceed a better solution we find. The optimal tour is found by finding the path that corresponds to the minimum lower bound.

Example: Traveling Salesman Problem

Find the optimal tour for a salesman visiting four cities (1, 2, 3, 4) with the following cost matrix.

$$C = \begin{pmatrix} \infty & 20 & 30 & 10 \\ 15 & \infty & 16 & 4 \\ 3 & 5 & \infty & 2 \\ 19 & 6 & 18 & \infty \end{pmatrix}$$

Step 1: Initial Lower Bound

We start by reducing the cost matrix to get the initial lower bound.

1. Row Reduction: Subtract the minimum value from each row.

Row 1 (min=10): $C1=(\infty 10 20 0)$

Row 2 (min=4): $C2=(11 \infty 12 0)$

Row 3 (min=2): $C3=(1 3 \infty 0)$

Row 4 (min=6): $C4=(13 0 12 \infty)$

Cost of Row Reduction: $10+4+2+6=22$.

Example: Traveling Salesman Problem

2. Column Reduction: Subtract the minimum value from each column of the row-reduced matrix.

- Column 1 (min=1): $(\infty 10012)^T$
- **Cost of Column Reduction:** 1. The other columns already have a zero, so no reduction is needed.

3. Total Lower Bound (LB): $22+1=23$.

The reduced matrix is: $C' =$

$$C' = \begin{pmatrix} \infty & 10 & 20 & 0 \\ 10 & \infty & 12 & 0 \\ 0 & 3 & \infty & 0 \\ 12 & 0 & 12 & \infty \end{pmatrix}$$

Example: Traveling Salesman Problem

Step 2: Branching from the Root

- We choose to explore the edge (3,1), which has a zero cost in the reduced matrix. To find the new lower bound for this branch, we apply the following rules:
- Add the cost of the original edge, $C(3,1)=3$.
- Add the lower bound from the parent node (the initial LB), which is 23.
- Remove row 3 and column 1.
- Remove the inverse edge (1,3) to prevent a sub-tour.

The new matrix is: $C''=$

$$C'' = \begin{pmatrix} \infty & 10 & \infty & 0 \\ 10 & \infty & 12 & 0 \\ \infty & \infty & \infty & \infty \\ 12 & 0 & 12 & \infty \end{pmatrix}$$

Example: Traveling Salesman Problem

We now perform a row/column reduction on this sub-matrix.

- Row 1 (min=0) -> no change
- Row 2 (min=0) -> no change
- Row 4 (min=0) -> no change
- Column 2 (min=0) -> no change
- Column 3 (min=12) -> new min=0: (12 -> 0), (12 -> 0).

Lower bound of sub-matrix: 12. New lower bound for branch (3,1):
 $23+0+12=35$.

Example: Traveling Salesman Problem

Step 3: Continuing the Branch

We are now in the state (3,1). Let's choose the next edge to explore, (4,2), which has a zero cost in our reduced sub-matrix.

1. Add the original cost $C(4,2)=6$.
2. Add the parent LB, which is 35.
3. Remove row 4 and column 2.
4. Remove inverse edge (2,4) to prevent a sub-tour.

The remaining matrix for the next two steps is: $C''' =$

$$\begin{pmatrix} \infty & \infty & 20 & 0 \\ 10 & \infty & 12 & \infty \\ \infty & \infty & \infty & \infty \\ \infty & \infty & 12 & \infty \end{pmatrix}$$

Example: Traveling Salesman Problem

There is only one full path left. The costs are:

- (1,4): $C(1,4)=10$ (from original matrix)
- (2,3): $C(2,3)=16$ (from original matrix)

This gives a total path of 3 -> 1 -> 4 -> 2 -> 3 with a cost of $3+10+6+16=35$.

Step 4: Final Solution

The algorithm would continue to explore other branches (e.g., edges with non-zero costs) and prune them if their lower bound exceeds 35. Since we found a complete tour with a total cost of 35, and it is equal to the lower bound calculated for this path, this is the optimal solution.

Optimal Tour: 3 -> 1 -> 4 -> 2 -> 3 Minimum Cost: 35

Lower Bound Theory

Lower bound theory concept is based upon the calculation of minimum time or maximum number of comparisons that is required to execute an algorithm is known as a lower bound theory or Base Bound Theory.

Concept/Aim: The main aim is to calculate a minimum number of comparisons required to execute an algorithm.